I'm Going to Stop, Will You Cross?: The Impact of Vehicle's eHMI on Pedestrian's Decision-Making

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Abstract. In SAE levels 4 and 5 autonomous vehicles (AVs), drivers are no longer required to control the vehicle, which may limit communication with pedestrians. To address this, external human-machine interfaces (eHMI) have been introduced as a means of facilitating interaction between AVs and pedestrians. The advisory and directive types of eHMI messages have been reported as effective in clearly conveying specific actions to pedestrians. However, these messages pose legal and ethical challenges and may result in various risks depending on the surrounding context. Therefore, this study analyzed the impact of eHMI messages conveying the movement intention of AVs on pedestrians' crossing decisions. A virtual reality experiment was conducted with 40 participants to evaluate pedestrians' crossing experiences based on eHMI message types and AV braking styles. Results indicated a tendency for movement information-based eHMI messages to reduce the pedestrian crossing decision time (CDT), although the difference was not statistically significant. However, a significant difference was observed in the CDT depending on the AV's braking style. These findings suggested that although eHMI messages may partially contribute to pedestrians' crossing decisions, the primary determinant remains the physical movement of the AV.

Keywords: eHMI · autonomous vehicles · movement information · crossing decision time

1 Introduction

1.1 Research Background

With the accelerated adoption of SAE levels 4 and 5 autonomous vehicles (AVs) that can operate without driver control or in the absence of a driver, situations may arise where communication between pedestrians and drivers is limited. In traditional road environments, pedestrians often encounter situations where they must understand the drivers' intentions to make safe crossing decisions. When drivers control vehicles, pedestrians can rely on nonverbal signals, such as eye contact, facial expressions, and head or

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H. Krömker (Ed.): HCII 2025, LNCS 15818, pp. 181–193, 2025.
https://doi.org/10.1007/978-3-031-92692-1_11

body movements, to communicate. Such interactions provide pedestrians with a sense of security [1]. However, in SAE levels 4 and 5 AVs, communication with drivers is no longer possible, making external human-machine interfaces (eHMI) a key technology for facilitating interactions between pedestrians and AVs [2]. Through interactions with AVs equipped with eHMI, pedestrians can enhance predictability, make quicker crossing decisions, feel a stronger sense of safety, and develop trust in the AVs [3, 4].

1.2 Limitations of Existing eHMI Approaches

Previous studies have categorized eHMI information into four types: awareness, intention, advisory, and commanding [5]. Awareness information conveys to pedestrians that the vehicle has detected them, whereas intention information helps pedestrians predict the vehicle's next movement. Advisory and commanding information suggest or direct specific actions for pedestrians to take. Research on the effects of different eHMI information types indicates that advisory messages are more intuitive than messages that merely convey the AV's status, and it is recommended to use short and commanding text to enhance message accuracy [6]. However, although advisory and commanding messages offer the advantage of clear and simplified communication with pedestrians, they are limited in real-world applications due to legal and ethical concerns [5, 7, 8]. For instance, messages such as "safe to cross" are known as intuitive eHMI messages that pedestrians can easily understand. However, such messages may be misinterpreted in complex environments or cause confusion among unintended pedestrians [5]. This demonstrates the potential risks associated with eHMI messages that advise or direct pedestrians toward specific actions. Moreover, pedestrians tend to rely on the vehicle's physical movements and their previous crossing experiences (e.g., habitual behaviors) rather than following directive eHMI messages [9–12]. These findings suggest that pedestrians do not passively accept eHMI information; instead, they integrate the AV's physical movements and eHMI messages when making crossing decisions.

1.3 Research Objectives and Contributions

Due to the limitations of advisory and commanding eHMI messages, an alternative approach has been proposed that conveys the vehicle's next movement intentions based on movement information. Movement information-based eHMI provide messages reflecting the vehicle's speed changes, acceleration, and deceleration states, and pedestrians exhibit a tendency to trust such information [5]. In particular, pedestrians tend to combine their previous crossing experiences with the physical movement states of the vehicle, such as its current speed, when making crossing decisions [12–15]. This underscores the critical role of vehicle speed information in pedestrians' crossing decisions, offering important implications for the eHMI design. Moreover, when eHMI messages are inconsistent with the vehicle's movements, pedestrians are less likely to blindly trust the eHMI and instead rely on the vehicle's physical movements [16]. This indicates that pedestrians place greater trust in and reliance on movement information when making crossing decisions. In addition, movement information aligns naturally with pedestrians' previous crossing experiences with human-driven vehicles, making it easier for pedestrians to accept eHMI technology [17]. Given the limitations of advisory and commanding eHMI

messages, this study designed eHMI messages based on vehicle movement information, considering pedestrians' previous crossing experiences. This study aimed to investigate the impact of movement information-based eHMI messages on pedestrians' crossing decisions.

2 Methodology

2.1 Participants

The experiment involved a virtual reality (VR) environment scenario where the participants experienced crossing in front of an AV from a pedestrian's perspective. As such, no qualifications, such as a driver's license or driving experience, were required for participation. However, individuals with visual or auditory impairments that could limit their ability to experience VR stimuli were excluded due to the use of VR equipment. In addition, the participants wearing corrective eyeglasses were excluded to ensure smooth VR headset use. The participants were recruited through university bulletin boards and social media platforms. Forty participants (25 males and 15 females), aged 20 to 53 years [mean (M) = 29.63 years; standard deviation (SD) = 6.34 years], took part in the experiment. The experiment lasted ~ 2 h, and the participants were compensated with 30,000 KRW (approximately $20) for their time. This study was conducted with approval from the Yonsei University Institutional Review Board (approval no. 7001988–202411-HR-2510–02) (Table 1).

Table 1. Demographic information of the participants

Participant information	Details
Number of the participants	40
Age	Mean = 29.63; SD = 6.34
Gender	25 males and 15 females

2.2 Experimental Design

Independent Variables

In this study, the independent variables were the eHMI type and AV braking style, and an experiment was designed to evaluate pedestrians' crossing experiences based on these variables. The eHMI types were divided into four categories and presented through the front display of the AV in a VR environment. The eHMI display was mounted at the center of the AV's front bumper, a location known to enhance message visibility, clarity, safety, and reduce cognitive load [18–20]. The first eHMI type was the Speed (km/h) eHMI, which displayed the AV's real-time speed in increments of 1 km/h. In addition, the vehicle's speed was dynamically designed to fluctuate within ± 1 km/h of the designated speed in the experiment. This dynamic eHMI was implemented to

prevent the driving speed from being continuously displayed as a static number, ensuring a more realistic representation. Such dynamic eHMI messages can lead pedestrians to perceive a higher level of perceived safety [21, 22]. The second eHMI type displayed the AV's longitudinal movement state using text. To enhance clarity, a single short word was selected for the message [23]. The longitudinal movement states were categorized into two types: driving, which indicates that the AV is in motion, and braking, which indicates that the AV is in a deceleration state. The term "braking" was used to represent the deceleration state rather than "stopping," which implies a complete stop. However, because the participants were nonnative English speakers and to improve the visibility of the text on the limited size of the eHMI display panel, the short and simple term "braking" was selected. To ensure that the participants did not misinterpret the meaning of the text eHMI in a way inconsistent with the study's intent, sufficient explanation was provided during the pre-experiment briefing. The third eHMI type combined the first and second types, simultaneously displaying Speed + Text information. Finally, a Baseline (No eHMI) condition was set, in which no movement information was provided on the eHMI display. This was used to compare pedestrians' crossing experiences with and without eHMI. All information displayed on the eHMI was presented in cyan. Providing eHMI information in specific colors, such as red or green, can result in unclear meanings or confusion for the participants [24]. Therefore, cyan (turquoise), which is known to maximize pedestrians' visibility, distinguishability, and perceived safety, was applied as the most suitable color for AV identification [25, 26] (Fig. 1).

Independent variables		eHMI type (4)			
		No eHMI	Speed	Text	Speed + Text
Braking style (2)	Soft				
	Hard				

Fig. 1. Experimental Conditions

Dependent Variables

In this study, the Crossing decision time (CDT) and Subject assessment measures were collected to evaluate the impact of AV eHMI types and braking styles on pedestrians' crossing experiences. The CDT was measured as the time elapsed from the moment the AV's eHMI was provided to the moment the pedestrian decided whether to cross. This metric was used to evaluate the decision-making speed of pedestrians when encountering an AV with eHMI in an approaching scenario. The participants were instructed to input their crossing decisions using a controller in a unity-based VR environment. By utilizing a controller integrated with the VR environment, it was possible to precisely collect timing data at the exact moment the participants, in the role of a pedestrian, decided whether to cross. Subject assessment measures were collected through a survey conducted after experiencing each crossing scenario under various conditions. The survey included evaluation metrics such as Understanding, Perceived safety, Mental workload, and Perceived

effort, which were collected using a seven-point Likert scale. The Understanding metric assessed how well the participants understood the AV's movement intentions based on its physical movements and eHMI information. Perceived safety reflected the pedestrians' subjective sense of safety regarding the approaching AV. The Mental workload and Perceived effort metrics indicated the cognitive load and effort required to make a crossing decision while observing the approaching AV. The CDT metric focused on quantitatively evaluating pedestrian behavior, whereas the Subject assessment measures were used to assess the psychological responses of pedestrians to the eHMI information.

2.3 Crossing Scenario

Experimental Environment

The experiment was conducted in a laboratory at Yonsei University. The pedestrian crossing scenario was developed using the Unity 3D simulation software to create a VR urban environment. To provide the participants with an immersive experience, the HTC VIVE Pro VR (a head-mounted display and controller) was used. The road surface on which the AV operated in the VR urban setting was designed as a shared space, resembling an open area rather than a typical road. To enable the direct observation of interactions between pedestrians and the eHMI-equipped AV, traffic elements such as crosswalks, traffic lights, and road lanes were excluded. This setup allowed pedestrians to make crossing decisions solely based on the approaching AV's movements and eHMI information without interference from the surrounding road conditions or traffic signals. In addition, the participants were instructed to indicate their crossing decisions using a controller to capture the moment of decision precisely rather than physically walking toward the AV. Specifically, the participants were asked to press the controller button when they felt they would "take the first step toward the approaching AV." The participants were guided to observe the AV's movements and eHMI information and press the controller button when they judged it as "safe to cross." Conversely, they were informed that if they could not determine a safe crossing moment, they were not required to press the button. This approach allowed the pedestrian's crossing decision-making process to be simulated in a manner similar to real-world road environments. Conducting the experiment in a VR environment rather than on an actual road ensured that data collection occurred in a consistent setting, free from external factors, while enabling repeated trials under controlled conditions for each participant (Fig. 2).

Composition of the Crossing Scenarios

In this study, the independent variables were the eHMI type (four conditions) and braking style (two conditions), resulting in eight distinct conditions. In addition, the scenario included two cases based on whether the AV stopped. The first case was when the AV decelerated and came to a complete stop at a specific point, and the second case was when the AV decelerated but continued driving at a low speed in front of the pedestrian. Including the stopping behavior of the AV in the scenario conditions minimized learning effects by preventing the participants from predicting that the AV would stop at specific points. Therefore, the crossing scenarios experienced by the participants were designed by combining the eight independent variable conditions with the two stopping conditions, resulting in 16 experimental conditions. A within-subject design was adopted to

Fig. 2. VR experimental setup for the pedestrian crossing scenario

ensure that all participants experienced every experimental condition. In addition, a Latin square design was applied to minimize order effects and learning effects. As a result, the 16 experimental conditions were randomly presented to the participants and evenly distributed across all conditions.

AV Driving Condition Settings

In the pedestrian crossing scenario, the driving conditions of the AV were categorized into two scenarios, Stop and Nonstop, as shown in Table 2. For each scenario, two braking styles, Soft and Hard braking, were set depending on the braking style. In all scenarios, the AV started at a point > 100 m away from the pedestrian, with an initial speed consistently set at 20 km/h. In the Stop scenario, for Soft braking, the vehicle began decelerating at a distance of 30 m from the pedestrian and was set to stop with a deceleration rate of − 1.0 m/s^2. In contrast, for Hard braking, the vehicle started decelerating at a distance of 20 m and stopped with a relatively strong deceleration rate of − 1.5 m/s^2. The final stopping point of the AV was set to be ~ 10 m away from the pedestrian. In the Nonstop scenario, the AV was configured not to stop in front of the pedestrian but to decelerate and proceed at a low speed. For Soft braking, the vehicle started decelerating at 30 m and reached a low speed of 3 km/h with a deceleration rate of − 0.8 m/s^2. For Hard braking, the vehicle began decelerating at 20 m and reached the low speed of 3 km/h with a relatively strong deceleration rate of − 1.3 m/s^2.

2.4 Experimental Procedure

The experiment was designed as a scenario to evaluate pedestrians' crossing experiences when encountering an AV equipped with movement information eHMI in a VR environment. First, the participants were provided with an explanation of the study's purpose and experimental procedure, followed by signing an informed consent form to participate in the study. Subsequently, the participants wore VR headsets and experienced a

Table 2. AV driving condition settings

Scenario case	Stop		Nonstop	
Braking style	Soft	Hard	Soft	Hard
Starting point (m)	100			
Starting speed (km/h)	20			
Deceleration start point (m)	30	20	30	20
Deceleration rate (m/s^2)	− 1	− 1.5	− 0.8	− 1.3
Slow speed (km/h)	—		3	
Stopping point (m)	10		—	

practice scenario to familiarize themselves with how to operate the controller to make crossing decisions from a pedestrian's perspective in the VR environment. In the main experiment, the participants experienced 16 conditions. These conditions consisted of eight combinations of four eHMI types and two braking styles, further divided into two cases based on whether the AV stopped after braking. All participants experienced all conditions following a within-subject design, and the order of scenarios was randomized using a Latin square design to minimize learning effects between conditions. At the start of each scenario, the AV began at a point ~ 100 m away from the participant at a speed of 20 km/h. The participants observed the AV as it approached a point ~ 50 m from them. They were instructed to determine the moment they judged it as "safe to cross in front of the AV. The participants expressed their crossing decision by pressing the controller button when they determined it was safe to cross. Conversely, if they judged it unsafe to cross, they did not press the button. After each scenario, the participants completed a subjective evaluation of their crossing experiences under the given condition using a seven-point Likert scale survey. After the completion of all scenarios, the participants were asked to respond to a post-experiment survey about their overall experience (Fig. 3).

3 Results

3.1 CDT Analysis by eHMI Type and Braking Style

For the collected pedestrian CDT data, the decision times of the participants who hesitated for an extended period or chose not to cross were recorded as NaN values. These NaN values were replaced with the maximum decision time recorded across all conditions for the respective participant. Instead of adding a fixed percentage to the maximum time, the exact maximum decision time was used to strictly account for crossing decision failures in the analysis (Table 3).

The mean CDT and 95% confidence intervals for each eHMI type are presented in Fig. 4. The mean (SD) CDT for the Baseline condition (No eHMI) was 3.22 (1.26) s, indicating a tendency for longer decision times compared to other groups. The mean (SD) CDT for the Speed eHMI type was 3.14 (1.18) s, showing a tendency to be shorter than the Baseline. The mean (SD) CDT for the Text and Speed + Text eHMI types was

Fig. 3. Crossing scenario in a VR environment.

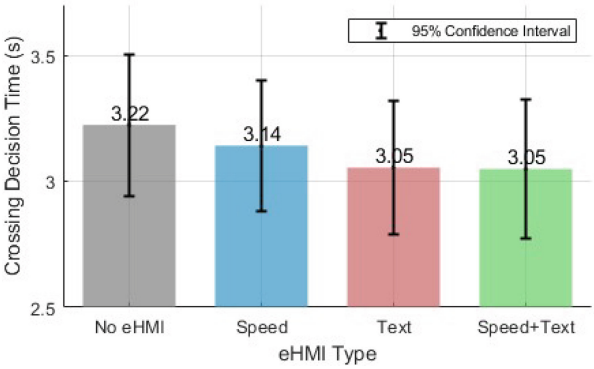

Fig. 4. Comparison of the Mean CDT by eHMI types

Table 3. Descriptive Statistics of the CDT by eHMI types

eHMI type	CDT (s)			
	Mean (s)	SD (s)	n	95% CI
No eHMI	3.22	1.26	80	2.94–3.50
Speed type	3.14	1.18	80	2.88–3.40
Text type	3.05	1.20	80	2.79–3.32
Speed + Text type	3.05	1.23	80	2.78–3.32

95% CI, 95% confidence interval.

3.05 (1.20) and 3.05 (1.23) s, respectively, which were the shortest CDTs among all groups (Table 4).

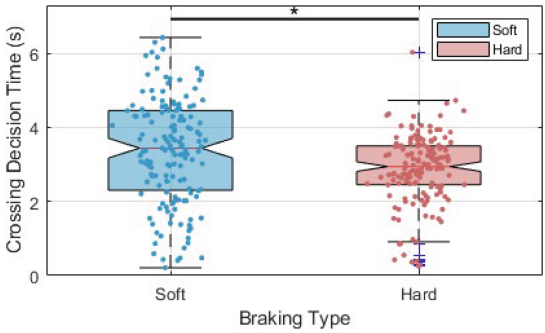

Fig. 5. Comparison of the Mean CDT by braking types

Table 4. Descriptive Statistics of the CDT by braking types

Braking type	CDT (s)			
	Mean (s)	SD (s)	n	95% CI
Soft	3.36	1.40	160	3.14–3.58
Hard	2.87	0.93	160	2.73–3.02

The results of pedestrians' CDT by braking type are shown in Fig. 5. The mean (SD) CDT for the Soft braking type was 3.36 (1.40) s, which was longer than the mean CDT for the Hard braking type, 2.87 (0.93) s, indicating that AVs with Soft braking require pedestrians to take more time to make crossing decisions compared to Hard braking. The results of the repeated-measures ANOVA for the four eHMI types revealed no statistically significant differences in the CDT between groups ($F(3, 117) = 1.641$, $p = 0.667$), indicating that the mean differences in the CDT across eHMI types were small, and eHMI information did not have a significant impact on pedestrians' CDT. In contrast, the AV's braking style had a significant effect on pedestrians' CDT ($F(1, 39) = 19.043$, $p < 0.001$), indicating that the CDT was significantly shorter with Hard braking compared to Soft braking. No statistically significant interaction effect was observed between eHMI type and braking style ($F(3, 117) = 1.342$, $p = 0.565$) (Fig. 6).

3.2 Subjective Evaluation Measures Analysis

The subjective evaluation measures were compared based on the participants' survey responses for the four eHMI conditions: No eHMI, Speed eHMI, Text eHMI, and Speed + Text eHMI, using one-way ANOVA. Welch's test was applied for the analysis, and in cases of significant differences, the Games-Howell non-equal variance multiple comparison test was conducted. For the participants' understanding of the AV's movement

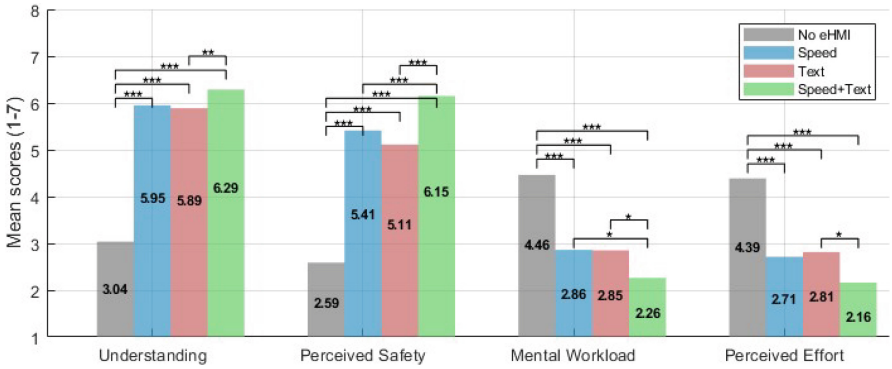

Fig. 6. Comparison of mean scores across conditions for each dependent variable. Statistical significance is indicated by asterisks: *p < 0.05, **p < 0.01, *p < 0.001.

intentions, a significant difference was observed among the eHMI types (F(3, 171.35) = 79.92, p < .001). Post hoc analysis revealed that the No eHMI condition (M = 3.04, SD = 1.72) had significantly lower scores compared to Speed eHMI (M = 5.59, SD = 1.02), Text eHMI (M = 5.89, SD = 0.81), and Speed + Text eHMI (M = 6.29, SD = 0.77) (all have a p-value < .001). Similarly, for Perceived Safety, a significant difference was found among the eHMI types (F(3, 170.03) = 101.33, p < .001). Post hoc analysis showed that the No eHMI condition (M = 2.59, SD = 1.61) had significantly lower perceived safety scores compared to Speed eHMI (M = 5.41, SD = 1.38), Text eHMI (M = 5.11, SD = 1.42), and Speed + Text eHMI (M = 6.15, SD = 0.87) (all have a p-value < .001). These results indicate that the inclusion of speed and text information in eHMI messages contributed to pedestrians' comprehension of the AV's movement intentions and enhanced their perceived safety. For Mental Workload, a significant difference was observed among the eHMI types (F(3, 174.47) = 29.03, p < .001). Post hoc analysis revealed that the No eHMI condition (M = 4.46, SD = 1.69) had significantly higher mental workload scores compared to Speed eHMI (M = 2.86, SD = 1.52), Text eHMI (M = 2.85, SD = 1.37), and Speed + Text eHMI (M = 2.26, SD = 1.28) (all have a p-value < .001). Additionally, the Speed + Text eHMI condition resulted in significantly lower mental workload compared to Speed eHMI (p = .038) and Text eHMI (p = .029). For Perceived Effort, a significant difference was found among the eHMI types (F(3, 174.63) = 25.52, p < .001). Post hoc analysis indicated that the No eHMI condition (M = 4.39, SD = 1.85) had significantly higher perceived effort scores compared to Speed eHMI (M = 2.71, SD = 1.52), Text eHMI (M = 2.81, SD = 1.57), and Speed + Text eHMI (M = 2.16, SD = 1.35) (all have a p-value < .001). Additionally, the Speed + Text eHMI condition showed significantly lower perceived effort than the Text eHMI condition (p = .028). These findings suggest that an eHMI message combining speed and text information effectively reduced pedestrians' mental workload and perceived effort during interactions with the AV.

4 Discussion

This study analyzed the effects of eHMI information based on the AV's physical movement and braking style on pedestrians' crossing experiences. The results showed that the impact of eHMI on pedestrians' CDT was limited, whereas the AV's physical movement, specifically braking style, had a significant influence on the CDT. Although eHMI information provided additional helpful cues for pedestrians making crossing decisions, pedestrians tended to rely on changes in the vehicle's physical movements to make their judgments. Under the experimental conditions of this study, hard braking clearly conveyed the vehicle's intention to stop, enabling pedestrians to make quicker decisions. In contrast, soft braking did not clearly reveal the vehicle's intention to stop, resulting in longer decision times for pedestrians. The Subjective evaluation measures included understanding of the AV's movement intention, perceived safety of the eHMI-equipped AV, mental workload during the crossing decision process, and perceived effort. The understanding of the AV's movement intention and perceived safety were highest when the AV was equipped with the Speed + Text eHMI, indicating that combining speed and text information effectively helped pedestrians understand the AV's movement intentions. In contrast, the No eHMI condition recorded the lowest scores across all measures, suggesting that the absence of visual information increased uncertainty in pedestrians' crossing decisions, leading to higher mental workload and perceived effort. These findings highlighted the importance of incorporating the AV's physical movement information, a key factor in pedestrians' crossing decisions, when designing eHMI. In particular, applying visual elements such as intuitive light-band patterns or road projection designs that reflect the AV's acceleration and deceleration could effectively enhance pedestrians' understanding of the AV's movement intentions. As this study was conducted under limited experimental conditions and focused on specific types of movement-based eHMI, further research considering a broader range of factors is necessary. Pedestrians' behaviors toward the AV's movement intentions may vary depending on age, individual crossing patterns, and driving experience. Future studies developing various eHMI designs based on physical movement information could contribute to improving pedestrians' crossing experiences with AVs.

Acknowledgments. This work was supported by the Korea Evaluation Institute of Industrial Technology grant funded by the Korean government (no. 20024368; Development of Integrated Control System for Multi-mobility and Autonomous Delivery Service Demonstration). This work was financially supported by the Korean Ministry of Land, Infrastructure and Transport as an "Innovative Talent Education Program for Smart City." We thank Taewoong Kim for his assistance with the experiment.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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